

DERIVED EQUIVALENCES OF MIRROR CALABI–YAU ORBIFOLDS AND SPECTRAL DEGENERATION

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ABSTRACT. We construct a canonical family of Fourier–Mukai transforms establishing triangulated equivalences between bounded derived categories of coherent sheaves on smooth Calabi–Yau orbifolds and the derived Fukaya categories of their symplectic mirrors. By examining the spectral sequence associated with the canonical filtration on Hochschild homology, we prove that the degeneration of the Hodge-to-de Rham spectral sequence at the E_1 -page is preserved under these derived equivalences. As an application, we demonstrate the stability of stringy topological invariants across mirror pairs under finite group actions, resolving a conjecture regarding mirror symmetry for quotient singularities in low dimensions.

1. INTRODUCTION

Homological mirror symmetry, originally formulated by Kontsevich [4], posits an intimate duality between the algebraic geometry of a Calabi–Yau variety X and the symplectic topology of its mirror dual Y . Mathematically, this duality is expressed as an equivalence of triangulated categories:

$$\Phi: \mathrm{D}^b\mathrm{Coh}(X) \xrightarrow{\sim} \mathrm{D}^b\mathrm{Fuk}(Y), \quad (1.1)$$

where $\mathrm{D}^b\mathrm{Coh}(X)$ denotes the bounded derived category of coherent sheaves on X , and $\mathrm{D}^b\mathrm{Fuk}(Y)$ represents the derived Fukaya category of Lagrangian submanifolds on Y .

When X admits the structure of a global quotient orbifold $\mathcal{X} = [X/G]$ by a finite group $G \subset \mathrm{SL}(n, \mathbb{C})$, the category of G -equivariant coherent sheaves $\mathrm{D}_G^b(X) = \mathrm{D}^b\mathrm{Coh}([X/G])$ serves as the appropriate substitute for $\mathrm{D}^b\mathrm{Coh}(X)$. In this context, the mirror dual object $\mathcal{Y}$ acquires non-trivial flat bundles or fractional branes, complicating the structural analysis of the Fukaya category.

The main purpose of this paper is to investigate the behaviour of spectral sequences governing the deformation theory and Hochschild homology of these categories.

1.1. Statement of Main Results. Our first primary result establishes the existence of an explicit equivalence for a broad class of quotient orbifolds.

Theorem 1.1. *Let X be a smooth projective Calabi–Yau threefold, and let $G \subset \mathrm{Aut}(X)$ be a finite abelian group acting faithfully and preserving the holomorphic volume form. Then there exists a mirror symplectic manifold $\mathcal{Y}$ equipped with a canonical fibration such that:*

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